

As part of a project exploring deep research techniques, the clinic team built an MCP server that allows agents access to novel data sources, including graph databases, to provide richer responses to user queries.

As a demonstration, students created a server that would allow an agent controlled through a chatbot the ability to query a knowledge graph of authors, papers, and topics. The graph shows who wrote what, which topics papers cover, and how papers cite each other. That lets users ask questions that simple search cannot handle, such as how two researchers are connected through their work or which authors are emerging in each area. At the bottom of the page is an example demo.

By the end of the project, the graph held a large collection of papers and authors, and the assistant could answer these relationship-based questions through a chat interface. The design is modular so new tools can be added without changing the core system, making it easy for Argonne to extend the assistant over time.

Compare trends between attention mechanisms and recurrent neural networks. Keep it very succinct.



Anthropic: Claude Sonnet 4.5

Based on the research data Argonne_compare...rch_topics :

Attention Mechanisms:

- 78 papers total (2021-2026)
- Declined 58.3% in publications
- Citations dropped from avg 190 (2021) to 1 (2026)

Recurrent Neural Networks:

- 62 papers total (2021-2026)
- Declined 82.4% in publications (steeper drop)
- Citations fell from avg 186 (2021) to 0 (2026)

Key Insight: Both topics are declining, but RNNs are fading faster. Attention mechanisms maintain 1.3x more research activity overall Argonne_compare...rch_topics .